B.Tech. Mathematics & Computing

COURSE PLAN: THEORY COURSE

School Name	**School of Basic Sciences, Humanities and Management**				
Course Name	**Artificial Intelligence**				
Course Code	**MAT 3132**	**Semester**	**V**	**Core / PE / OE**	**Core**
Name of the faculty	**Dr Aishwarya S.**				
No of contact hours/week:		**L**	**T**	**P**	**C**
		2	**1**	**0**	**3**

Lesson Plan

L No	Topics	CO Addressed
0	*To address OBE philosophy towards NBA and IET accreditation followed by a discussion on Course Plan, Course Outcomes, CO-PO Mapping and Blooms Taxonomy, with prerequisites for the course, if any.*	
1	AI Agents and rationality; Task environments and architecture types	CO1
2	Search problems and examples; Best-first search	CO1
3	Uninformed search: Uniform-cost search, BFS, DFS	CO1
4	Tutorial	CO1
5	Informed search: Greedy, A*	CO1
6	Local search: Hill climbing, Simulated annealing, Genetic algorithms	CO1
7	Tutorial	CO1
8	Adversarial search: Minimax, Alpha-beta pruning	CO1
9	Constraint Satisfaction Problems: Backtracking, Arc consistency	CO1
10	Tutorial	CO1
11	Logic-based representations: PL and FOL	CO1
12	Inference with propositional logic: Forward/backward chaining; Brief introduction to Prolog	CO1
13	Tutorial	CO1
14	Probabilistic reasoning and uncertainty	CO2
15	Bayes nets and reasoning with them	CO2
16	Tutorial	CO2
17	Hidden Markov Models	CO2
18	EM algorithm, Mixture of Gaussians, Anomaly detection	CO2
19	Tutorial	CO2
20	Supervised learning: Decision trees	CO2
21	Naive Bayes Classifier	CO2
22	Tutorial	CO2
23	Linear and logistic regression, Gradient descent	CO2
24	Neural networks: Architecture, Backpropagation	CO2
25	Overview of CNNs and RNNs	CO2
26	Nearest Neighbours; Unsupervised learning: k-Means Clustering	CO2
27	Reinforcement learning: MDPs, Concept of Return, Model-based RL	CO3
28	Tutorial	CO3
29	Value iteration, Policy iteration	CO3
30	Model-free RL: Q-learning	CO3
31	Tutorial	CO3
32	NLP: N-gram models, smoothing, language models	CO3
33	NLP: Vector representations; Introduction to attention and transformers	CO3
34	Tutorial	CO3
35	Applications to computer vision	CO3
36	Applications to robotics	CO3

Self Directed Learning (SDL)

Sl. No.	TOPIC	No. of learning hours	CO
1	Depth-limited search, Iterative deepening search, Backtracking search, Iterative Deepening A* (IDA*) search	8	CO1
2	Inference in first-order logic	8	CO1
3	Dimension reduction techniques	8	CO2
4	Kernel functions	8	CO2
5	Active and passive reinforcement learning	8	CO3
6	Basics of image formation, Object detection and segmentation	8	CO3
7	Bias and fairness in AI, AI safety and alignment, AI ethics	6	CO3
	Total number of learning hours	**54**	

***The SDL topics may be from the Research Literature/ Case Studies NPTEL/Coursera/any MOOC-based material Virtual Lab, etc.**

****One credit is equivalent to 30 hours of learning. (4 credits course =120 hrs learning ; 48+72=120 hrs of learning)**

References:

Textbooks	1. Stuart Russell and Peter Norvig: Artificial Intelligence - A Modern Approach (4th Edition), Pearson (2020).
	2. Tom Mitchell: Machine Learning, McGaw Hill (1997).
	3. Kevin P.Murphy: Machine Learning – A Probabilistic Perspective, MIT Press (2012).
	4. C.M.Bishop: Pattern Recognition and Machine Learning, Springer (2006).
	5. R.O.Duda, P.E.Hart, D.G.Stork: Pattern Classification, John Wiley & Sons (2006).
	6. T. Hastie, R. Tibshirani, J. Friedman: The Elements of Statistical Learning – Data Mining, Inference and Prediction, Springer (2009).
	7. David J.C. MacKay: Information Theory, Inference, and Learning Algorithms, Cambridge University Press (2003).
	8. Q. Yang: Intelligent Planning – A Decomposition and Abstraction Based Approach, Springer (1997).
Self-Directed Learning	
Research Literature/ Case Studies	
NPTEL/Coursera/any MOOC-based material	
Virtual Labs	

Submitted by: Dr Aishwarya S.